DISCRETE TRANSISTOR OP-AMP

Jason Gallicchio and Claude, rev A, 2026-10-24

github.com/gallicchio/op-amp-pcb
circuit, scripts, schematic, this board

The op-amp of Learning the Art of Electronics
(Hayes & Horowitz, 2016), built from eight
2N3904 / 2N3906 so you can probe every node.

POWER: +15 V, GND, −15 V into the terminal, 20 mA.
  Both green LEDs light when it is right way round.
SWITCHES, usual: TAIL 1 on (Q6 current source),
  COMPENSATION 1 on (100 pF + 1k), FEEDBACK as needed.
  TAIL 2 = 10k tail resistor instead.
  COMPENSATION 2 shorts the 1k: the book's bare
  100 pF; the x11 amp still works, a follower rings.
FEEDBACK 1 = X to IN−: gain 11. 2 = OUT to IN−:
  follower. Both off: it is a bare op-amp.
AS AN OP-AMP IN YOUR OWN CIRCUIT (Labs 6-9): FEEDBACK
  both off; IN−, IN+, OUT and GND on the terminals
  are pins 2, 3, 6 of a '411. Mind its 2 uA of input
  bias current (no 10 M feedback); its output limits
  at 20 mA, like a real one (Q7, Q8: not in the book).
OFFSET NULL: as a follower with IN+ grounded,
  turn RV1 until OUT reads 0 V.
INPUTS: ±12 V. OUT: ±13 V into 1k or more; a short
  to ground or a rail is survived, not enjoyed.
PROBING: a loop on every named node for a scope
  hook, and GND loops for the clip.

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